EXPERIMENT 18

TITLE: WRITE A PROLOG PROGRAM FOR A DATABASE WITH NAME AND DOB

Aim

To develop a Prolog program to create a simple database containing the names and dates of birth of persons and retrieve the required information using queries.

Objective

- To understand the concept of facts and predicates in Prolog.
- To create a simple database using Prolog.
- To store names and dates of birth.
- To retrieve information using Prolog queries.
- To understand logical querying in Prolog.

Theory

A database in Prolog can be represented using **facts**. Each fact describes a relationship between objects or values. In this experiment, the dob/2 predicate is used to store the name and date of birth of a person.

For example, the fact:

dob(varshini, '22-12-2007').

represents that Varshini's date of birth is 22-12-2007.

Prolog can retrieve stored information by using queries. Variables can also be used to find the name of a person when the date of birth is given.

Algorithm

Step 1: Start the program.

Step 2: Define the dob/2 predicate.

Step 3: Store the names and corresponding dates of birth as facts.

Step 4: Use a query to find the date of birth of a particular person.

Step 5: Use a variable to find the person corresponding to a given date of birth.

Step 6: Display the retrieved information.

Step 7: Stop the program.

Flowchart

START

```
        |
        ▼
  Create DOB Database
        |
        ▼
  Store Name and DOB
        |
        ▼
   Enter a Query
        |
        ▼
  Search the Database
        |
        ▼
  Is Matching Fact Found?
     /        \
    Yes        No
     |         |
     ▼         ▼
 Display Result   Display "No Match"
     |         |
     └─────────┬─────────┘
        |
        ▼
       STOP
```

Prolog Program

% Database containing name and date of birth

dob(varshini, '22-12-2007').

dob(rahul, '15-03-2006').

dob(priya, '10-07-2007').

dob(arun, '25-01-2006').

dob(kavya, '18-11-2007').

% Rule to find the date of birth of a person

find_dob(Name, DOB) :-

    dob(Name, DOB).

Queries

Query 1: Find the DOB of Varshini

?- find_dob(varshini, DOB).

Sample Output

DOB = '22-12-2007'.

Query 2: Find the person having DOB 15-03-2006

?- find_dob(Name, '15-03-2006').

Sample Output

Name = rahul.

Query 3: Display all names and their DOBs

?- dob(Name, DOB).

Sample Output

Name = varshini,

DOB = '22-12-2007' ;

Name = rahul,

DOB = '15-03-2006' ;

Name = priya,

DOB = '10-07-2007' ;

Name = arun,

DOB = '25-01-2006' ;

Name = kavya,

DOB = '18-11-2007' ;

false.

Explanation

The predicate dob(Name, DOB) represents the relationship between a person's name and date of birth.

For example:

dob(varshini, '22-12-2007').

means that Varshini's date of birth is 22-12-2007.

The rule:

```
find_dob(Name, DOB) :-
   dob(Name, DOB).
```

allows the user to retrieve the stored information through queries.

When a variable is used in a query, Prolog searches the database and returns all matching facts.

Advantages

- Simple way to represent structured information.
- Easy to retrieve information using queries.
- Supports logical reasoning.
- Multiple facts can be stored and searched efficiently.
- Demonstrates the basic concept of a knowledge base in Prolog.

Applications

- Personal information databases.
- Student record systems.
- Employee databases.
- Knowledge representation systems.

- Expert systems.
- Artificial Intelligence applications.

Result

The Prolog program was successfully implemented to create a database containing names and dates of birth. The required information was successfully retrieved using Prolog queries.

Conclusion

Thus, a simple Name and Date of Birth database was successfully implemented using Prolog. This experiment demonstrates how facts, predicates, rules, variables, and queries can be used for storing and retrieving information in a Prolog-based knowledge system.

```
PS C:\Users\Varshini M\AppData\Lo
!!!/ARTIFICIAL INTELLIGENCE/VSC E
1. Find DOB using Name
2. Find Name using DOB
3. Display All Records
4. Exit
Enter your choice: 1
Enter Name: Arun
Name: Arun
Date of Birth: 15-03-2005

--- NAME AND DOB DATABASE ---
1. Find DOB using Name
2. Find Name using DOB
3. Display All Records
4. Exit
Enter your choice: 
```

```python
# Database with Name and Date of Birth

database = {
    "Arun": "15-03-2005",
    "Priya": "22-07-2004",
    "Rahul": "10-01-2006",
    "Divya": "18-11-2005",
    "Karthik": "05-09-2004"
}


# Find DOB using Name
def find_dob(name):
    if name in database:
        print("Name:", name)
        print("Date of Birth:", database[name])
    else:
        print("Person not found.")


# Find Name using DOB
def find_name(dob):
    found = False

    for name, date in database.items():
        if date == dob:
            print("Date of Birth:", dob)
            print("Name:", name)
            found = True

    if not found:
        print("Person not found.")


# Display complete database
def display_database():
    print("\nName\t\tDate of Birth")
    print("--------------------------------")

    for name, dob in database.items():
        print(name, "\t\t", dob)


# Menu
while True:
    print("\n--- NAME AND DOB DATABASE ---")
    print("1. Find DOB using Name")
    print("2. Find Name using DOB")
    print("3. Display All Records")
    print("4. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        name = input("Enter Name: ")
        find_dob(name)

    elif choice == "2":
        dob = input("Enter Date of Birth (DD-MM-YYYY): ")
        find_name(dob)

    elif choice == "3":
        display_database()

    elif choice == "4":
        print("Program terminated.")
        break

    else:
        print("Invalid choice.")
```